



Biology H Review Practice

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Range: Chapter 12-13

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

I. MCQ (60', 1Q = 3')

1. In Griffith's experiment, all of the following would not cause mice to die EXCEPT:
A) heat-killed S strain
B) heat-killed S strain and living R strain
C) living R strain
D) heat-killed R strain
2. Avery proved DNA is genetic material because transformation failed in the test tube with:
A) Proteinase
B) ribonuclease
C) deoxyribonuclease
D) ligase
3. If Hershey-Chase found *only* S-35 radioactivity in bacteria, it would mean:
A) Protein entered bacteria
B) DNA contains genetic material
C) Viruses lack DNA
D) RNA carried genetic information
4. Chargaff's rules show that in DNA:
A) $A + G = T + C$
B) $A = T$ and $G = C$
C) Purines > Pyrimidines
D) Bases absorb visible light
5. Rosalind Franklin's X-ray diffraction revealed DNA's:
A) Nucleotide sequence
B) Double-helix with bases near center
C) Telomere function
D) Transcription process
6. Topoisomerase prevents issues during replication by:
A) Separating the double-helix structure
B) Relieving overwinding strain
C) Synthesizing primers
D) Joining fragments
7. Telomerase inactivity in somatic cells causes:
A) Limited cell division
B) Faster aging
C) More mutations

- D) Crossing over
8. Alternative RNA splicing enables:
- A) Faster translation
 - B) The production of multiple proteins from one gene
 - C) Intron retention
 - D) DNA replication
9. tRNA with anticodon *AUU* pairs with mRNA codon:
- A) ATG
 - B) UAA
 - C) none
 - D) UAC
10. Sickle cell anemia's Glu→Val mutation is a:
- A) Silent mutation
 - B) Nonsense mutation
 - C) Missense mutation
 - D) Frameshift mutation
11. Prokaryotic transcription/translation overlap due to:
- A) Circular DNA
 - B) The lack of nucleus
 - C) Stable mRNA
 - D) Small ribosomes
12. Translocation causes CML(Chronic myelogenous leukemia) by:
- A) Deleting genes
 - B) Activating uncontrolled cell division gene
 - C) Inactivating repair
 - D) Duplicating chromosomes
13. During DNA replication, DNA polymerase III without proofreading causes:
- A) Faster replication
 - B) More mutations
 - C) Telomere loss
 - D) Primer failure
14. Polyploidy benefits plants by making them:
- A) Disease-resistant
 - B) Larger/stronger
 - C) Tastier
 - D) Insect-resistant

15. Retroviruses violate central dogma by:
- A) DNA → protein
 - B) RNA → DNA
 - C) Protein → RNA
 - D) RNA → protein
16. Eukaryotic RNA polymerase binds promoters via:
- A) Transcription factors
 - B) tRNA
 - C) Ribosomes
 - D) Primers
17. A nonsense mutation creates:
- A) Same amino acid
 - B) New amino acid
 - C) Stop codon → nonfunctional protein
 - D) Frameshift
18. Okazaki fragments joined by:
- A) Helicase
 - B) DNA pol I
 - C) DNA ligase
 - D) RNA Primase
19. In the Hershey-Chase experiment, bacteriophages were grown with radioactive phosphorus-32 (^{32}P) to label their DNA. After allowing these phages to infect bacteria, where was the ^{32}P radioactivity primarily detected?
- A) In the empty protein coats outside the bacteria
 - B) Inside the bacterial cells (host cytoplasm)
 - C) In newly assembled phage particles
 - D) In the liquid culture medium
20. RNA splicing keeps all of the following EXCEPT:
- A) Exons
 - B) Introns
 - C) Poly-A tails
 - D) 5' caps

II. FRQ (40')

1. A researcher compares DNA replication in *E. coli* (prokaryote) and human cells (eukaryote).
(a) Contrast TWO structural differences between prokaryotic and eukaryotic DNA.

- (b) Explain why the lagging strand requires discontinuous synthesis. Include:

- Role of Okazaki fragments
- Enzyme that joins fragments

- (c) Describe ONE consequence if DNA polymerase loses its proofreading function.

2. DNA template: 3'-TAC GTA CTT GCA-5'

- (a) Transcribe to mRNA and translate to amino acids using:

AUG = Methionine, CAU = Histidine, GAA = Glutamic acid, CGU = Arginine, CCA = Proline

- (b) A mutation changes the DNA to 3'-TAC GTA CTT CCA-5':

- Identify mutation type
- Compare original vs. mutated amino acid sequence

- (c) Why do eukaryotes process mRNA (cap, tail, splicing) before translation?

3. Cancer cells show abnormal DNA replication fidelity and telomerase activity.

- (a) Compare telomerase activity in cancer cells vs. somatic cells.

(b) Contrast the biological impact of polyploidy in plants vs. animals.
